Polytopality of mixes and their regular covers

Human Compression project

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Abstract

We apply the common fiber-intersection argument to the criteria of Cunningham–Hubard, in the complete author edition arXiv:2506.19334v1. The main criterion uses every flag and is proved without a finiteness assumption. Facet coverings, parity doubles and two-orbit regular covers follow from the same test. We also exhibit a counterexample to the printed coprime-label sufficient condition: the eleven-cell and the four-simplex have a nonpolytopal mix. Several printed single-base conclusions require correction. A second explicit construction explains this failure, while the all-flags theorem and its precisely rooted consequences remain valid. This working edition is awaiting independent scientific review.

1 The main criterion and its premises

Use the notation of the common paper. For two rooted components R_1, R_2 with flag stabilizers N_1, N_2 in their universal connection group, write

$$X(R_2 \mid R_1) = N_2 N_1 / N_2$$

when N_1 normalizes N_2. This is the *lower* variance group, with the order of the arguments shown explicitly. It is not merely an abstract group type: for the facet, vertex-figure and medial groups below we use the embeddings into $\mathrm{Aut}(P_2)$ proved in common Lemma 4.

Theorem 1. *Let P_1 be an n-polytope and P_2 an n-premaniplex. Suppose both are admissible over a common rooted quotient T. For a flag $q = (\Psi_1, \Psi_2)$ of their mix let K_i, L_i, N_i denote its rooted facet, vertex-figure and medial section in the ith factor. Then $P_1 \diamond P_2$ is polytopal if and only if its facets and vertex-figures are polytopal and, at every such flag,*

$$X(K_2 \mid K_1) \cap X(L_2 \mid L_1) \leq X(N_2 \mid N_1). \tag{1}$$

Proof. The common quotient supplies a subgroup normalizing and containing both flag stabilizers. Common Lemma 4 identifies the three variance groups with the three fibers in common (6). Common Corollary 3 then gives the assertion. All identifications are made at the specified flag. This is [CH, Theorem 6.3], including the necessity in its Proposition 6.2. □

The restricted facets and vertex-figures are themselves rooted mixes. Thus when all relevant rooted facet mixes and vertex-figure mixes are polytopal, a trivial intersection on the left of (1) suffices, giving the form of [CH, Corollary 6.4] with its flag choices explicit. Common Proposition 6 proves [CH, Proposition 6.5]: a facet covering at every paired flag forces polytopality. Its proof includes the required induction on the vertex-figures, not just vanishing of one variance group.

When the common quotient remains connected on middle colors, common Proposition 5 permits one base flag in (1). More generally one needs only one representative per middle-color component of that quotient. In particular there is one test for regular inputs and for two-orbit inputs whose common two-vertex quotient has an edge of an interior color. These are conclusions of an actual common symmetry action on the mix. We have not replaced that hypothesis by separate transitivity assertions about its two factors.

2 Why separate medial transitivity does not give one test

The common-quotient condition in the preceding section cannot simply be replaced by medial-section transitivity of the two inputs separately. We give two isomorphic polytopes with different compatible rootings that satisfy all the local and symmetry hypotheses in [CH, Corollaries 6.4, 6.6, Theorem 6.7 and Proposition 6.8]. The indicated base test passes, although their mix is not polytopal. In particular, this is not a counterexample depending on nonuniform facets or on a nonpolytopal input.

2.1 A six-coordinate action

Let $S = S_5$, with $s_i = (i+1\ i+2)$ for $0 \le i \le 3$. Its six Sylow 5-subgroups are indexed $0, \dots, 5$ using the respective cycle generators

$$(12345),\ (12354),\ (12453),\ (12435),\ (12543),\ (12534).$$

Conjugation induces a permutation action π on these six coordinates. Writing each image list in order $0, \dots, 5$, the four permutations are

i	$\pi(s_i)(0)$	$\pi(s_i)(1)$	$\pi(s_i)(2)$	$\pi(s_i)(3)$	$\pi(s_i)(4)$	$\pi(s_i)(5)$
0	4	2	1	5	0	3
1	5	3	4	1	2	0
2	3	2	1	0	5	4
3	1	0	4	5	2	3

(S1)

Use the induced left permutation action on $V = \{v \in \mathbb{F}_3^6 : \sum_h v_h = 0\}$, so $(gv)_h = v_{\pi(g)^{-1}h}$. In $V \rtimes S$ put $R_i = (u_i, s_i)$, where

$$u_0 = u_2 = u_3 = 0, \qquad u_1 = (-1, -1, 0, 1, 0, 1).$$

Multiplication is $(u, g)(v, h) = (u + gv, gh)$. Direct substitution in (S1) shows $R_i^2 = 1$ and $[R_i, R_j] = 1$ for $|i - j| > 1$; also

$$(R_0R_1)^3 = (R_2R_3)^3 = 1, \qquad c := (R_1R_2)^3 = (v, 1), \quad v = (1, 1, -1, -1, 1, -1). \tag{S2}$$

These identities are small coordinate calculations, all displayed inputs being fixed elements of $\mathbb{F}_3$ and S_5.

The four marks generate all of $G = V \rtimes S$. Indeed their projection is S, and their normal closure of c contains the S-span of v. The five vectors $v, s_0v, s_3v, s_0s_3v, s_1s_0v$ have rank five over $\mathbb{F}_3$; their rows are

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 1 & 2 \\ 1 & 2 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 2 & 2 & 2 \\ 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 2 & 1 & 2 & 1 & 1 \end{pmatrix}.$$

They lie in V, so span V. Consequently $|G| = 3^5 \cdot 120 = 29160$.

For $h \in \{0, \dots, 5\}$ let $N_h = \{v \in V : v_h = 0\}$ and form the colored Schreier graph $P_h = N_h \backslash G$, with adjacency given by right multiplication by R_i. Using right rather than left connection actions merely reverses the names of words; the colored path statements are unchanged. A flag can be written $(g, k) \in S \times \mathbb{F}_3$, and its i-adjacency is

$$(g, k)^i = (gs_i, k + (gu_i)_h). \tag{S3}$$

Thus every P_h has 360 flags and covers the simplex flag graph $T = S$. The cover is regular with deck group $V/N_h \cong C_3$. Equivalently, the preimages of $N_h \trianglelefteq V$ in the universal connection group satisfy exactly the source's definition of T-admissibility. Since T is a maniplex, so are these covers.

2.2 Every input is a medial-section-transitive polytope

Write $A = \langle R_0, R_1, R_2 \rangle$, $B = \langle R_1, R_2, R_3 \rangle$ and $C = \langle R_1, R_2 \rangle$. Their translation kernels under $G \to S$ are

$$\begin{aligned} V_A &= \langle e_0 - e_5, e_1 - e_2, e_3 - e_4 \rangle_{\mathbb{F}_3}, \\ V_B &= \langle e_0 - e_3, e_1 - e_5, e_2 - e_4 \rangle_{\mathbb{F}_3}, \\ V_C &= \langle v \rangle_{\mathbb{F}_3}. \end{aligned} \tag{S4}$$

Here is a full justification of these kernel formulas. The projected end groups are S_4, with their usual adjacent-transposition presentations. All its Coxeter relations already hold in each lifted end except the middle cube in (S2). Therefore its kernel is generated by the conjugates of that cube, and equals the linear span of the corresponding orbit of v. The first space in (S4) is invariant under s_0, s_1, s_2, contains v, and the three vectors $v, s_0 v, s_1 s_0 v$ are independent in it. The second space is invariant under s_1, s_2, s_3, and contains the independent vectors $v, s_3 v, s_2 s_3 v$. This proves both equalities. The last follows also from the presentation of a group generated by two involutions: $R_1 R_2$ has order nine and its projection has order three. We retain here only the elementary Coxeter presentations of symmetric groups; the translation kernels are computed in full above.

The restricted groups on colors $\{0, 1\}$ and $\{2, 3\}$ project isomorphically to S_3, by (S2). Consequently each facet of P_h covers a tetrahedron with trivial fiber on its $\{0, 1\}$ sections. The common fiber criterion gives the rank-three intersection property: one of the two fibers being intersected is trivial. Its two-color sections are polygons, so every facet is polytopal. The dual argument proves the same for every vertex-figure. Each facet and vertex-figure has $24 \cdot 3 = 72$ flags, since every coordinate projection of both spaces in (S4) is surjective. Every medial section has $6 \cdot 3 = 18$ flags, since all coordinates of v are nonzero. At every flag of P_h the medial fiber over T is therefore the whole C_3 fiber. The common criterion now proves that P_h itself is polytopal. This proves all-flag assertions, not only a test at a selected root.

The stabilizer H_h of the hth Sylow subgroup in S_5 is its normalizer, an affine group of order 20 on five points. To see this directly, identify the five points with $\mathbb{F}_5$: translations normalize the chosen regular C_5, as do the four nonzero scalings, and a normalizer is determined by the image of one point and of a cycle generator. This group is sharply two-transitive. Since $V \rtimes H_h$ normalizes N_h, its left action on $N_h \backslash G$ commutes with all colored adjacencies. On T, a middle-color component is specified by the ordered pair $(g(1), g(5))$. There are twenty such pairs, and H_h is transitive on them. Each of these components has one connected preimage in P_h, as just proved. Hence P_h is medial-section-transitive. In particular its facets, vertex-figures and medial sections have uniform isomorphism types. The medial components themselves form one automorphism orbit, exactly as required by the source. The six inputs are isomorphic: conjugation by a suitable $(0, a)$ sends N_h to N_k, and left multiplication by $(0, a)$ then gives a color-preserving map $N_h \backslash G \to N_k \backslash G$.

2.3 One base passes and another fails

Consider P_0 and P_1, rooted at compatible flags specified below. Their mix is $(N_0 \cap N_1) \backslash G$, with 1080 flags. Its flags are (g, k, l), and (S3) holds in both sheet coordinates. All its facets and vertex-figures are polytopal, since they are mixes of polyhedra.

At a flag above $g \in S$, the three fibers of this mix over T are the images of gV_A, gV_B, gV_C under the two coordinate projections (v_0, v_1). This follows by traversing a word whose projection returns to g in the corresponding color subgroup. At $g = 1$, (S4) gives

$$D_A = D_B = \mathbb{F}_3^2, \qquad D_C = \langle (1, 1) \rangle.$$

Thus the end intersection fails: it has 54 flags whereas the middle component has 18. There is also an explicit path witness. Put $h = R_2 R_3$ and use the translation c from (S2). The words

$$(cR_0 cR_0)^2 \in A, \qquad (chc^{-1}h^{-1})^2 \in B$$

both carry $(1,0,0)$ to $(1,1,0)$, by (S1)–(S2). No middle-color word can do so, since the middle translation fiber is the diagonal line. The mix is consequently not polytopal.

Now set $g_* = s_1 s_2 s_3 = (2\,3\,4\,5)$, and use base flags with $(g,k,l) = (g_*,0,0)$. From (S1), $\pi(g_*) = (4,1,0,2,3,5)$. The coordinate pair $(0,1)$ of $g_* z$ is therefore (z_2, z_1). Equations (S4) give at this base

$$D_A = \langle(-1,1)\rangle, \qquad D_B = \mathbb{F}_3^2, \qquad D_C = \langle(-1,1)\rangle. \tag{S5}$$

The base intersection thus holds. To express the test exactly as a mix of P_0 with P_1, rather than a cover of T, require that the first sheet remain fixed. Intersect the three fibers in (S5) with $\{0\} \times \mathbb{F}_3$. The facet and middle fibers become trivial; the vertex fiber is C_3. In particular the two end variance groups have trivial intersection at the specified base.

Here and throughout this paper, medial-section transitivity means transitivity on the middle-color components, as used in the proof of [CH, Proposition 2.5] and its two-orbit tests. The parenthetical chain type on the source's physical page 6 should instead be $\{0, n-1\}$; it prints the complementary type. We do not silently conflate these notions.

Both inputs are polytopes and individually medial-section-transitive, both are admissible over the same regular T, all their local types are uniform, and even every mixed end is polytopal. Nevertheless the base conditions hold while the mix fails elsewhere. This disproves the single-base conclusions of [CH, Corollary 6.4 and Theorem 6.7] as printed, and of Corollary 6.6 and Proposition 6.8 under the operational medial-section convention just specified. The source's all-flags Theorem 6.3 is unaffected. A correct replacement either keeps every flag, assumes medial-section transitivity of the mix itself, or uses the explicitly proved common-quotient condition $W = CV$. The same argument identifies exactly what went wrong: separate automorphisms of the two factors need not act together on their chosen mix.

3 Polyhedra and coprime middle labels

The mix of any two rooted polygons is a polygon: its connected flag graph is again a simple two-colored cycle or double ray. In fact the mix of any two polyhedra is polytopal, without a symmetry assumption. To prove this, apply the common fiber criterion at colors $\{0,1\}$ and $\{1,2\}$. If a point of the common fiber belongs to both, projection to the second polyhedron places it either at y or at $r_1 y$. The latter is impossible. A word in r_0, r_1 fixing the first polygon flag has even length, whereas a word carrying the second polygon flag to its 1-adjacent flag has odd length. Both assertions follow from the bipartite flag graph of a polygon. Thus the fiber intersection is trivial, as required. This supplies the input cited from earlier work in [CH, Proposition 6.8]. For rank four, therefore, condition (1) is the only remaining test once the appropriate flag representatives have been justified.

Proposition 2. *Let P_i be orientable regular four-polytopes of types $\{p_i, q_i, r_i\}$, where q_1, q_2 are finite and coprime. Then $P_1 \diamond P_2$ is polytopal.*

Proof. In the second medial dihedral group D_{2q_2}, the lower variance group of a q_2-gon relative to a q_1-gon is the full rotation subgroup: $\langle(\rho_1\rho_2)^{q_1}\rangle = \langle\rho_1\rho_2\rangle$. The intersection of the two end variance groups lies in $\langle\rho_0,\rho_1,\rho_2\rangle \cap \langle\rho_1,\rho_2,\rho_3\rangle = \langle\rho_1,\rho_2\rangle$, because P_2 is polytopal. Every word fixing a first-factor facet flag has even length, since P_1 is orientable. Its image in the second group is therefore even under the orientation character of P_2. Consequently the common variance group lies in the rotation subgroup of the medial dihedral group. Theorem 1 and the polyhedron result prove the claim. It is enough in this proof that one first-factor end be orientable, while the second polytope is orientable. □

The orientability assumption on the first factor cannot be dropped as it is in the literal [CH, Proposition 6.9]. The following is a complete finite obstruction, using the familiar eleven-cell. Its marked group and intersection property are already established in our projective-group paper; a short standalone finite certificate is being bound here so that the cost of that input is visible.

Let P_1 have the following involutions on eleven points:

$$\begin{aligned} a_0 &= (1\,2)(3\,5)(4\,7)(6\,8), \\ a_1 &= (1\,3)(2\,4)(5\,7)(6\,9), \\ a_2 &= (3\,6)(4\,7)(5\,8)(9\,10), \\ a_3 &= (1\,3)(2\,5)(4\,7)(10\,11). \end{aligned} \tag{2}$$

These generate the simple group of order 660, with a C-marking of type $\{3, 5, 3\}$. Both end groups are A_5 and their intersection is the middle dihedral group of order ten. Let P_2 be the four-simplex, with $b_i = (i+1\ i+2)$ in S_5. It is orientable, via the sign character, and its middle label is three, coprime to five.

The mixed facet projects onto A_5 and S_4. A subdirect product of these groups is their full direct product: by the elementary common quotient description of a subdirect product, a proper one would give a nontrivial common quotient; simplicity of A_5 and $60 > 24$ exclude this. The same argument applies to the other end. Their intersection in the product group is consequently

$$D_{10} \times S_3,$$

of order 60. The common marked middle is generated by two involutions whose product has order $\operatorname{lcm}(5, 3) = 15$, so is dihedral of order 30. More explicitly, $(a_1, 1)$ belongs to both ends. It does not belong to the common marked middle, since in that middle the reflection character of D_{10} equals the sign character of S_3, whereas their values on $(a_1, 1)$ differ. Thus the mix fails the intersection property. Both inputs are regular polytopes, the second is orientable, and the middle labels are coprime, exactly as required in the printed proposition. The defect in its proof is that evenness of the first-factor relations was inferred from orientability of the second factor.

4 A common quotient faithful on a medial section

Suppose P_1, P_2 are regular n-polytopes with facets K_1, K_2. Suppose both facets cover a regular premaniplex K_3, and the covering from K_2 is an isomorphism on its vertex-figures. This is equivalent to the source's isomorphism-type hypothesis: two isomorphic regular sections have a base-preserving isomorphism, obtained by composing with an automorphism, and a covering preserving that base is uniquely determined by colored paths. Thus the given covering is that isomorphism, also for infinite sections. Then their mix is polytopal, as in [CH, Proposition 6.14]. Here is the full kernel argument. Every relation of K_1 maps trivially to K_3, so $X(K_2 \mid K_1)$ lies in the kernel of $\mathrm{Aut}(K_2) \to \mathrm{Aut}(K_3)$. That kernel meets the base vertex-figure subgroup trivially. The intersection with $X(L_2 \mid L_1)$ lies in that very subgroup, by the intersection property of P_2, and is consequently trivial.

There are still the local polytopality obligations. The vertex-figures of K_1 cover those of K_3, hence those of K_2. The dual of the common facet-cover proposition makes $K_1 \diamond K_2$ polytopal. Likewise the facets of the vertex-figure L_1 cover those of L_2, so $L_1 \diamond L_2$ is polytopal. Theorem 1 finishes the argument. This proves the uniform statement; applying it to square or cubic toroids retains the classical fact that their translation quotient with unit period has the same spherical vertex-figure. We do not count the classification of regular toroids as a new consequence of this proof.

5 Two tests with their actual embeddings

For completeness we give the exceptional two-orbit reduction without suppressing the root of any variance group. This is the precise interpretation needed when applying [CH, Corollaries 6.10–6.12]. Let $V = \ker \epsilon_I$ contain and normalize both input stabilizers, and suppose every interior color belongs to I. Choose a paired base flag q_0 and the second representative $q_1 = r_j q_0$, where $j \notin I$. Write

$$A_k = X(K_2(q_k) \mid K_1(q_k)), \quad B_k = X(L_2(q_k) \mid L_1(q_k)), \quad C_k = X(N_2(q_k) \mid N_1(q_k)).$$

Each is embedded in the actual automorphism group of the second input using its displayed paired root. The exact criterion is local polytopality at the listed types and

$$A_k \cap B_k \le C_k \qquad (k = 0, 1). \tag{9}$$

The common orbit theorem proves sufficiency and necessity. The number of local types and the possible common subgroup are as follows.

I^c	j	facet types	vertex types	common subgroup
$\{0, n-1\}$	0	1	1	$A_0 = A_1$
$\{n-1\}$	$n-1$	2	1	$B_0 = B_1$
$\{0\}$	0	1	2	$A_0 = A_1$

Indeed a facet orbit corresponds to a component of the two-vertex quotient after deleting color $n-1$, and similarly for vertex orbits. This proves the type counts, including coincident isomorphic types. To justify the subgroup equalities, the global lower variance group acts on the mix by automorphisms which fix its first coordinate. Its subgroup stabilizing a mixed facet is exactly the facet variance group: an element preserves that facet if and only if it sends its base flag inside the same facet. The stabilizer is independent of the chosen flag inside that facet. Thus moving the base by r_0 leaves A_k unchanged. Moving it by r_{n-1} similarly leaves B_k unchanged. This proves every equality in the table.

The last two rows reproduce the two displayed tests of Corollaries 6.11 and 6.12 with their embeddings explicit. In the first row, (9) reads $A_0 \cap B_0 \le C_0$ and $A_0 \cap B_1 \le C_1$. Even when both vertex-figures have the same isomorphism type, their embeddings must be transported before identifying B_0 and B_1. If an actual common embedding makes them equal, these two tests become $A_0 \cap B_0 \le C_0 \cap C_1$, the expression printed in Corollary 6.10. The table (9) is the unconditional replacement. We do not infer equality of embedded subgroups solely from equality of abstract rooted-section types, and do not claim a counterexample to the literal two-orbit version of Corollary 6.10 here.

6 Parity doubles

Let $P = W/N$ be a regular polytope. Fix a nonzero character $\epsilon : W \to C_2$ whose values on the generators prescribe the chosen flag coloring, and let Q be the mix of P with the two-flag quotient $W/\ker\epsilon$. Its stabilizer is $N \cap \ker\epsilon$. For every color subgroup S its lower variance subgroup in C_2 is exactly

$$E_S = \epsilon(N \cap S). \tag{3}$$

It is trivial precisely when the coloring descends to that rooted section of P. The character formulation also handles a restriction that is zero on all its colors: the corresponding connected quotient section is then a single flag.

Write A, B, C for the facet, vertex-figure and middle color subgroups. If $E_A = 1$ or $E_B = 1$, the corresponding facet or vertex-figure of P covers that section of the two-flag quotient. The common facet-cover proposition or its dual gives polytopality of Q. If $E_A = E_B = C_2$ and $E_C = 1$, (1) fails and Q is not polytopal. Finally, if $E_C = C_2$, (1) is automatic and Q is polytopal precisely when both mixed ends are polytopal. These are all three clauses of [CH, Proposition 6.13], with restricted trivial characters made explicit.

7 The smallest regular cover of a two-orbit polytope

Let $P = W/N$ be a two-orbit polytope of class 2_I. Put $V = \ker\epsilon_I$, so N is normal in V and $[W : V] = 2$. For $s \in W \setminus V$, the normal core of N in W is

$$N \cap sNs^{-1}. \tag{4}$$

Indeed all conjugates under V agree, and the remaining conjugates are sNs^{-1}. Hence the smallest regular premaniplex cover is the mix of P with its other rooting $\overline{P}$. This is a statement about the smallest regular *premaniplex* cover; whether it is polytopal remains the test (1), not an automatic consequence of (4).

Because (4) is normal in W, the mix $P \diamond \overline{P}$ is regular. The final observation in the common orbit theorem therefore gives a single-base criterion for *every* two-orbit class: its two rooted mixed ends must be polytopal, and their two lower variance groups must have intersection in the medial variance group at that same base. The extra automorphisms of this regular mix may interchange the two coordinate roles; the proof does not require them to project to automorphisms of either input.

This supplies all four class-dependent decision criteria in the subject of [CH, Corollary 6.15]. To identify the actual ends, choose the paired roots $(\Phi, s\Phi)$ in the underlying polytope P. If P is facet-transitive, its two facets have the same unrooted type K, and their rooted mix is $K \diamond \overline{K}$. If its facets have two orbits, the two roots pick the two types K, K', giving $K \diamond K'$. The analogous alternatives give $L \diamond \overline{L}$ or $L \diamond L'$ for the vertex-figures. A facet has two orbits only for $I^c = \{n-1\}$, and a vertex-figure has two orbits only for $I^c = \{0\}$, by the quotient components in the preceding table. The medial term is formed at the same paired root, so it is the variance of the two specified medial sections, including distinct types when $I^c \subseteq \{0, n-1\}$.

If one wishes to retain the source's two representative format, apply (9) to these two re-rootings. Regularity of the mix makes the two path tests equivalent, although the displayed coordinate embeddings can differ. The source's simplified intersection involving two medial variance groups must be read with those transport data. Our criterion keeps one actual paired base and makes no additional identification.

Here is the further simplification behind [CH, Proposition 6.16 and Corollary 6.17]. If $n-1 \notin I$, each $(n-2)$-face is regular. Given two flags of that face, extend them to flags of P. If they are in different global automorphism orbits, change the second extension in color $n-1$. This preserves the given lower face flag and changes its global orbit. An automorphism now maps the two extensions and restricts to the required face automorphism. If also $|I| < n-1$, the facets of P belong to one orbit: deleting the last color from the two-vertex quotient leaves an edge. They therefore have one type K, whose facets are regular. The mix $K \diamond \overline{K}$ has identical rooted regular facets in the two factors, so the common facet-cover proposition proves that it is polytopal.

When both extreme colors lie outside I, this applies to both ends. If in addition an interior color lies outside I, the one-base criterion therefore reduces to

$$X(K \mid \overline{K}) \cap X(L \mid \overline{L}) \leq X(N_0 \mid \overline{N_0}),$$

where N_0 now denotes the medial section, not a flag stabilizer. This recovers the chiral specialization and the stated two-orbit extension with the medial-orbit hypothesis explicitly discharged. The borderline case with no such interior color belongs to the two-test list above.

8 Replacing the second factor by an admissible cover

Suppose $N_1 \trianglelefteq V$ and the second stabilizer N_2 is normal in all of W. The mix $T \diamond P_2$ has stabilizer $V \cap N_2$, normal in V. Moreover

$$N_1 \cap (V \cap N_2) = N_1 \cap N_2.$$

Thus $P_1 \diamond P_2 = P_1 \diamond (T \diamond P_2)$ as rooted premaniplexes, and the latter is a mix of two T-admissible inputs. This proves the replacement in [CH, Section 6.6]. If the first input is polytopal, the main theorem applies without a polytopality requirement on the refined second factor. If only the second is known polytopal, one must separately establish polytopality of its refinement; the parity-double analysis shows why that obligation is real.

9 Current source boundary

The common paper and the preceding sections give complete written proofs of the all-flags criterion, the normalizer interpretation and the displayed consequences. The single-base formulations are replaced by the proved orbit condition and the explicit six-coordinate obstruction. The exceptional two-orbit rows retain their actual embedded subgroups as in (9); the stronger identification suppressed in Corollary 6.10 is not imported as a premise. The printed Proposition 6.9 has the explicit obstruction above and a corrected orientable sufficient condition.

The three questions in [CH, Section 6.6] are not claimed solved. No numerical factor or principal-source credit is attached to this working edition. The complete common theory is part of any eventual proof and measurement boundary, independently of its overlap with earlier topics.

References

[1] G. Cunningham and I. Hubard, *Polytopality criteria for the mix of polytopes and maniplexes*, arXiv:2506.19334v1, 24 June 2025, 21 physical pages.